

LM head (tied) — next token; slot_id logit masked

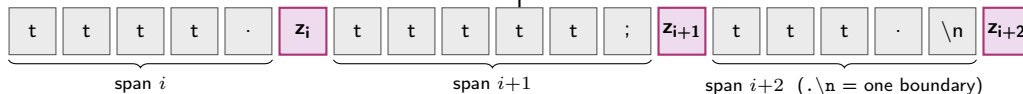
CODA (4 layers, ALL positions) — token input = input_norm(Prelude), dropped w.p. p [Bowman]; slot input = $h_i W_1, h_i W_2$
 tokens attend slots as ordinary positions (window + CSA/HCA) — the plan is a *prefix* the coda refines [Block Transformer; Optimus]
 prefix_k = 2: position 1 loss-free (the plan), position 2 emits; label = first token of span $i+1$ [BT App. F.2; MegaByte $p=0$]
 HC mean $\rightarrow$ lm_mixer $\rightarrow$ final_norm $\rightarrow$ fused CE

unchanged: HC carrier ($n=4$)
 GLA retention
 MORTAR / ReMoE
 ternary + int6 QAT
 AdEMAMix $\beta_1=0$
 TST phase 1 (plain MORPH)
 TUL on at the TST switch

CORE $\times T_i$ (6 shared layers) — **SLOT positions only**
 gather $\rightarrow$ loop $\rightarrow$ scatter; $T_i \sim \text{Poisson}(\text{mean_depth})$ per slot; masked update
 DiagonalInjection, GLA carry, TBPTT — as today
 h_i attends $h_{<i}$ every layer, every iteration [Coconut]
 h_i has **no loss of its own** [MegaByte, H-Net, LD4LG, Pred-Sent]

tokens: prelude state,
skip the core
 [BLT Eq. 9; MORPH seed path]

PRELUDE (4 layers, ALL positions, causal, window + CSA/HCA, GLA on layer 1)
 tokens: embed + bigram + value-embeds + $\times 0$ slots: pooled in context by attention over the span [BLT §3.2.2]



boundary rule $.;!?- \backslash n$ (no comma), min span 4, cap 32; slot input = $E_{\text{slot}} + \text{mean}(\text{embed of span } i)$ [TST; DTP; BLT Eq. 5]